ARTIN-SCHREIER EXTENSIONS: A Z/PZ GALOIS EXTENSION IN CHARACTERISTIC P IS GENERATED BY A ROOT OF X^P - X - C

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The chapter prefaces this lemma with the injective character G -> Z/pZ, sigma -> sigma(b) - b, and calls the Artin-Schreier theory a converse; it is not part of the counted unit.

Editorial note. The label lemma-Artin-Schreier names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-independence-characters (delivered card L48335), theorem-galois-theory (delivered card L48341, for the fixed-field step), lemma-finite-Galois (|G| = [L : K] = p for a finite Galois extension with group Z/pZ); freshman's dream in characteristic p

Source reference. Original reference 0CKL: sources/stacks-fields/pinned-original-source.tex, source lines 1676–1676. Prerequisite pointer invoked at 3560; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1 (Artin-Schreier extensions)**.** *Let L/K be a Galois extension of fields of characteristic $p > 0$ with Galois group $\mathbf{Z}/p\mathbf{Z}$. Then $L = K[z]$ with $z^p - z \in K$.*

Proof. Let σ be a generator of $\mathrm{Gal}(L/K)$. Consider $\sigma : L \to L$ as a K-linear operator. Observe that $\sigma^p - 1 = 0$ as a linear operator. Applying linear independence of characters (Lemma 0CKL), there cannot be a polynomial of degree $< p$ annihilating σ. We conclude that the minimal polynomial of σ is $x^p - 1 = (x - 1)^p$. This implies that there exists $w \in L$ such that $(\sigma - 1)^{p-1}(w) = y$ is nonzero. Then $\sigma(y) = y$, i.e., $y \in K$. Thus $z = y^{-1}(\sigma - 1)^{p-2}(w)$ satisfies $\sigma(z) = z + 1$. Since $z \notin K$ we have $L = K[z]$. Moreover since $\sigma(z^p - z) = (z + 1)^p - (z + 1) = z^p - z$ we see that $z^p - z \in K$ and the proof is complete. □

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